

A Global Structural Reduction for the Spherical-Shell Proxy

Global analytic structural module

Abstract

This note gives the global structural component of the purely analytic middle-ratio argument for the spherical-shell proxy. The weighted proxy is first written as an exact layer-cake sum. Its deficit from the action integral is then decomposed into a radial shifted-quadrature deficit and an angular rounding defect. A Stieltjes calculation retains two nonnegative remainders that are discarded by the optical estimate, and one of those remainders admits a smooth explicit lower bound. The result is the direct lower bound (36). All structural identities hold for every $0 < \rho < 1$ and $K > 0$; in particular they apply on the upward-closed middle-ratio region $\rho_c \leq \rho \leq 39/50$, $K \geq k_{11}(\rho)$.

1 Action and the global target

For $a > 0$ and $z \geq 0$, put

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \quad (1)$$

Fix $0 < \rho < 1$ and $K > 0$, and define

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z), \quad T := A(0) = \frac{(1-\rho)K}{\pi}. \quad (2)$$

The strict positive-integer count is

$$[x]_{<} := \#\{n \in \mathbb{N} : n < x\}.$$

The spherical-shell proxy and its continuous payment are

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu \left[A(\nu) + \frac{1}{4} \right]_{<}, \quad (3)$$

$$W(\rho, K) := \int_0^K 2z A(z) dz = \frac{2(1-\rho^3)K^3}{9\pi}. \quad (4)$$

The upward-closed middle-ratio region used below is

$$\mathcal{R}_c := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho) \right\}, \quad \rho_c := \frac{1}{1+2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}. \quad (5)$$

The middle-ratio target is $P < W$ on $\mathcal{R}_c$.

For $0 < z < K$, direct differentiation gives

$$-A'(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \frac{z}{K}, & \rho K \leq z < K. \end{cases} \quad (6)$$

In particular, A is strictly decreasing from T to 0. Let

$$R : [0, T] \longrightarrow [0, K]$$

be its decreasing inverse, and set

$$F(t) := R(t)^2, \quad g(t) := -F'(t), \quad \tau := A(\rho K) = KG_1(\rho). \quad (7)$$

The derivative in (6) increases in magnitude on $(0, \rho K)$ and decreases in magnitude on $(\rho K, K)$. The inverse curvature has the related, but slightly stronger, property

$$g \text{ decreases on } (0, \tau), \quad g \text{ increases on } (\tau, T). \quad (8)$$

Indeed, on the outer branch, if $z = R(t)$ then

$$g(t) = \frac{2\pi z}{\arccos(z/K)};$$

the quotient increases with z , while z decreases with t . On the inner branch, put

$$\delta(z) := \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}.$$

Differentiation with respect to the intermediate ratio gives

$$\frac{\delta(z)}{z} = \int_{\rho}^1 \frac{ds}{s\sqrt{s^2 K^2 - z^2}}.$$

This integral increases with z . Thus $z/\delta(z)$ decreases with z , and $g(t) = 2\pi z/\delta(z)$ increases as t increases and z decreases. The endpoint and interface data are

$$F(\tau) = \rho^2 K^2, \quad g(\tau) = \frac{2\pi \rho K}{\arccos \rho}, \quad g(T) = \frac{2\pi \rho K}{1 - \rho}. \quad (9)$$

Here and below $g(T)$ is the one-sided limit. Near $t = 0$, one has $g(t) = O(t^{-1/3})$, so all improper Stieltjes boundary terms used below vanish.

2 Exact layer cake and angular rounding

Let

$$x_n := n - \frac{1}{4}, \quad M(r) := \# \left\{ \ell \in \mathbb{N}_0 : \ell + \frac{1}{2} < r \right\} = \max \left\{ 0, \left\lceil r - \frac{1}{2} \right\rceil \right\}. \quad (10)$$

Lemma 1 (Exact layer cake). *For every $0 < \rho < 1$ and $K > 0$,*

$$P(\rho, K) = \sum_{\substack{n \geq 1 \\ x_n < T}} M(R(x_n))^2, \quad W(\rho, K) = \int_0^T F(t) dt. \quad (11)$$

Both identities retain the strict convention at every radial and angular wall.

Proof. For a fixed half-integer ν ,

$$\left[A(\nu) + \frac{1}{4} \right]_{<} = \#\{n \geq 1 : x_n < A(\nu)\}.$$

Since A and R are decreasing, $x_n < A(\nu)$ is equivalent to $\nu < R(x_n)$. Interchanging the two finite sums and using

$$\sum_{\ell=0}^{m-1} (2\ell + 1) = m^2$$

proves the first identity. At $R(x_n) = m + 1/2$, the strict angular count is m , exactly as specified by M .

For the second identity, the area formula for a decreasing inverse gives

$$\int_0^T R(t)^2 dt = \int_0^K 2zA(z) dz.$$

The last integral is (4). □

For every active n , abbreviate

$$r_n := R(x_n), \quad m_n := M(r_n), \quad \alpha_n := r_n - m_n + \frac{1}{2}. \quad (12)$$

The strict convention implies

$$m_n - \frac{1}{2} < r_n \leq m_n + \frac{1}{2}, \quad 0 < \alpha_n \leq 1. \quad (13)$$

This includes $m_n = 0$. Define the angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{x_n < T} (m_n^2 - r_n^2). \quad (14)$$

The defect of a single radial layer is exactly

$$m_n^2 - r_n^2 = (1 - 2\alpha_n)r_n + \left(\alpha_n - \frac{1}{2}\right)^2. \quad (15)$$

Unlike the one-sided estimate $m_n^2 - r_n^2 < r_n + 1/4$, formula (15) retains the cancellation when $\alpha_n > 1/2$.

Combining (11) and (14) gives the first exact two-term decomposition:

$$W - P = \mathcal{D}_{\text{rad}} - \mathcal{E}_{\text{ang}}, \quad \mathcal{D}_{\text{rad}} := \int_0^T F(t) dt - \sum_{x_n < T} F(x_n). \quad (16)$$

3 The radial deficit with retained positive remainders

Define the shifted counting function and its two primitives by

$$C(t) := \#\{n \geq 1 : x_n < t\}, \quad w(t) := t - C(t), \quad \mathfrak{W}(t) := \int_0^t w(s) ds, \quad \Psi(t) := \mathfrak{W}(t) - \frac{t}{4}. \quad (17)$$

On $0 \leq t \leq 3/4$,

$$\Psi(t) = \frac{1}{2} \left(t - \frac{1}{4} \right)^2 - \frac{1}{32}. \quad (18)$$

For $t = x_n + u$, $0 \leq u \leq 1$,

$$\Psi(t) = \frac{1}{2} \left(u - \frac{1}{2} \right)^2 - \frac{1}{32}. \quad (19)$$

Consequently

$$\mathfrak{W}(t) \geq 0, \quad -\frac{1}{32} \leq \Psi(t) \leq \frac{3}{32} \quad (t \geq 0). \quad (20)$$

Proposition 2 (Exact retained-remainder identity). *For every $0 < \rho < 1$ and $K > 0$, put*

$$\mathcal{R}_{\text{dec}} := - \int_{(0,\tau)} \mathfrak{W}(t) dg(t), \quad (21)$$

$$\mathcal{R}_{\text{osc}} := \left(\Psi(T) + \frac{1}{32} \right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi(t) \right) dg(t). \quad (22)$$

Then both remainders are nonnegative, and

$$\begin{aligned} \mathcal{D}_{\text{rad}} &= \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau) \left(\frac{\tau}{4} + \frac{3}{32} \right) \\ &\quad + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}}. \end{aligned} \quad (23)$$

Therefore

$$W - P = \mathcal{B}(\rho, K) + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}} - \mathcal{E}_{\text{ang}}, \quad (24)$$

where

$$\mathcal{B}(\rho, K) := \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4} + \frac{3}{32} \right). \quad (25)$$

Proof. Distributionally, $dw = dt - dC$. Stieltjes integration by parts, using $F(T) = 0$ and $w(0) = 0$, gives

$$\mathcal{D}_{\text{rad}} = \int_0^T w(t)g(t) dt. \quad (26)$$

The improper boundary term at 0 vanishes because $\mathfrak{W}(t) = t^2/2$ there and $g(t) = O(t^{-1/3})$.

On $(0, \tau)$, integrate with $d\mathfrak{W} = w dt$:

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_{(0,\tau)} \mathfrak{W} dg. \quad (27)$$

On (τ, T) , use $w = 1/4 + \Psi'$ and $\int_\tau^T g = F(\tau)$:

$$\int_\tau^T wg = \frac{F(\tau)}{4} + \Psi(T)g(T) - \Psi(\tau)g(\tau) - \int_{(\tau,T)} \Psi dg. \quad (28)$$

Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, adding (27) and (28) cancels the uncontrolled interface value $\Psi(\tau)$.

Now add and subtract the extremal sawtooth values in (20). The exact identity

$$\begin{aligned} &\Psi(T)g(T) - \int_{(\tau,T)} \Psi dg \\ &= -\frac{g(T)}{8} + \frac{3g(\tau)}{32} + \left(\Psi(T) + \frac{1}{32} \right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi \right) dg \end{aligned}$$

gives (23). By (8), $-dg$ is a positive measure on $(0, \tau)$ and dg is a positive measure on (τ, T) . Equations (20), (21), and (22) therefore prove nonnegativity of both remainders. Finally substitute (9) and $\tau = KG_1(\rho)$, and then use (16). $\square$

4 A smooth lower bound for the decreasing-branch remainder

The term $\mathcal{R}_{\text{dec}}$ is quantitatively important on the middle-ratio region; discarding it loses too much. It has the following elementary smooth minorant.

Lemma 3 (A global primitive minorant). *For $t \geq 0$, define*

$$L(t) := \frac{t^2}{2(1+2t)}, \quad L'(t) = \frac{t(1+t)}{(1+2t)^2}. \quad (29)$$

Then

$$0 \leq L(t) \leq \mathfrak{W}(t) \quad (t \geq 0). \quad (30)$$

Proof. For $0 \leq t \leq 3/4$, one has $\mathfrak{W}(t) = t^2/2 \geq L(t)$. For $t \geq 3/4$, (20) gives

$$\mathfrak{W}(t) \geq \frac{t}{4} - \frac{1}{32}.$$

Moreover,

$$\frac{t}{4} - \frac{1}{32} - \frac{t^2}{2(1+2t)} = \frac{6t-1}{32(1+2t)} \geq 0.$$

This proves (30). □

Define

$$\mathcal{R}_*(\rho, K) := - \int_{(0,\tau)} L(t) dg(t). \quad (31)$$

Because $-dg$ is positive on the decreasing branch,

$$0 \leq \mathcal{R}_* \leq \mathcal{R}_{\text{dec}}. \quad (32)$$

Integration by parts is legitimate at 0 because $L(t)g(t) = O(t^{5/3})$. Hence

$$\mathcal{R}_* = -L(\tau)g(\tau) + \int_0^\tau L'(t)g(t) dt. \quad (33)$$

On the outer branch $t = A(z)$, one has $g(t) dt = -2z dz$. Therefore

$$\mathcal{R}_* = -\frac{\tau^2}{2(1+2\tau)}g(\tau) + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz. \quad (34)$$

This expression is smooth at every point with $0 < \rho < 1$ and $K > 0$ and has no floor or wall discontinuity.

Corollary 4 (Direct retained-remainder bound). *For every $0 < \rho < 1$ and $K > 0$, define*

$$\begin{aligned} \mathcal{H}(\rho, K) := & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz, \quad \tau = KG_1(\rho). \end{aligned} \quad (35)$$

Then

$$W(\rho, K) - P(\rho, K) \geq \mathcal{H}(\rho, K) - \mathcal{E}_{\text{ang}}(\rho, K). \quad (36)$$

In particular, $\mathcal{E}_{\text{ang}}(\rho, K) < \mathcal{H}(\rho, K)$ implies $P(\rho, K) < W(\rho, K)$.

Proof. By (25), (34), and

$$\frac{\tau}{4} - \frac{\tau^2}{2(1+2\tau)} = \frac{\tau}{4(1+2\tau)},$$

one has $\mathcal{H} = \mathcal{B} + \mathcal{R}_*$. Now use (24), (32), and $\mathcal{R}_{\text{osc}} \geq 0$. □

Remark 5 (Handoff). The companion one-layer estimate proves, whenever at least one shifted radial layer is active,

$$\mathcal{E}_{\text{ang}} < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2}, \quad N = \#\{n \geq 1 : n - 1/4 < T\}.$$

Combining this with (36) gives one direct smooth lower bound for $W - P$. The separate double-sawtooth expansion is not needed for this implication and is kept outside the live proof as a research reformulation.